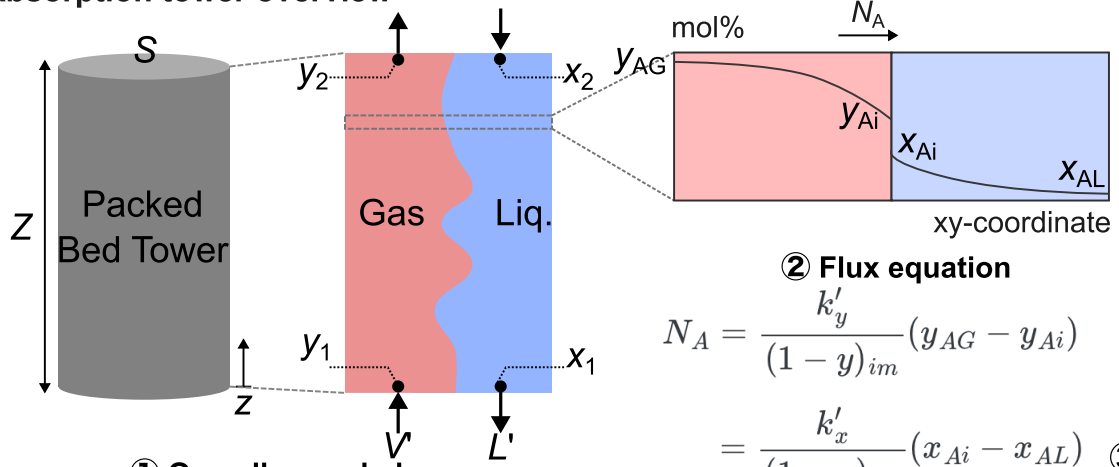
Cheatsheet: Absorption /Stripping Tower

Absorption tower overview



① Overall mass balance

$$L'\left(\frac{x_2}{1-x_2}\right)+V'\left(\frac{y_1}{1-y_1}\right)=L'\left(\frac{x_1}{1-x_1}\right)+V'\left(\frac{y_2}{1-y_2}\right)$$

Relations for flow rates

$$L = L' + L_x \qquad V_y = V\cdot y = V'\frac{y}{1-y}$$

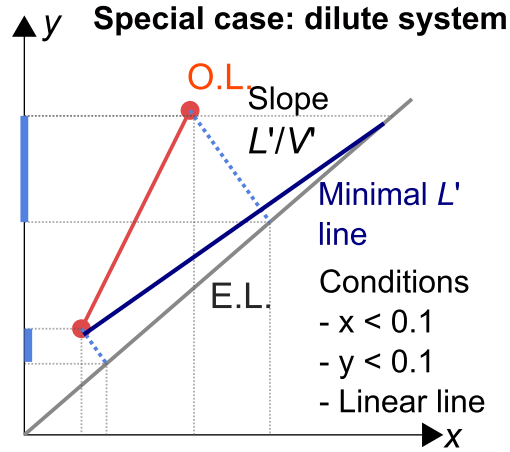
$$V = V' + V_y \qquad L_x = L\cdot x = L'\frac{x}{1-x}$$

② Flux equation

$$N_A = \frac{k'_y}{(1-y)_{im}}(y_{AG}-y_{Ai})$$

$$= \frac{k'_x}{(1-x)_{im}}(x_{Ai}-x_{AL})$$

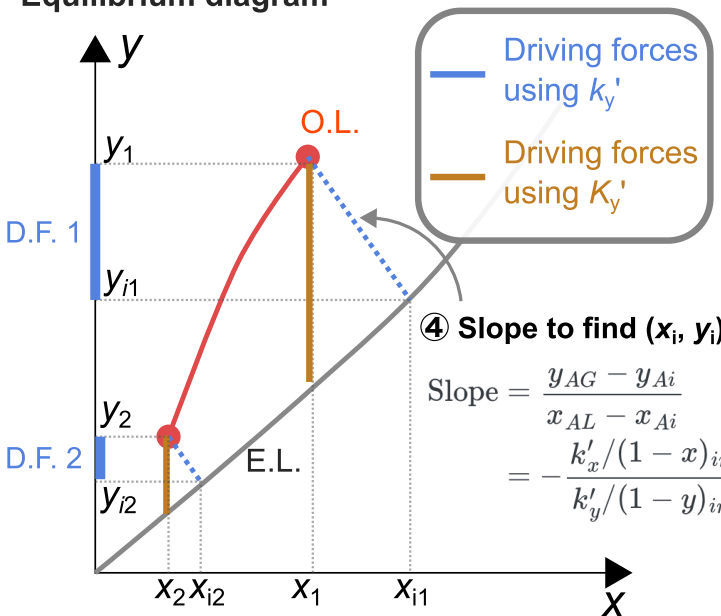
Special case: dilute system



③ Operating line slope (dilute)

$$y = \left(\frac{L'}{V'}\right)x + \left[y_1 - \left(\frac{L'}{V'}\right)x_1\right]$$

Equilibrium diagram



④ Slope to find (x_i, y_i)

$$\text{Slope} = \frac{y_{AG}-y_{Ai}}{x_{AL}-x_{Ai}} = -\frac{k'_x/(1-x)_{im}}{k'_y/(1-y)_{im}}$$

⑤ Height of tower

Gas

$$Z = \int_{y_2}^{y_1} \frac{V\,dy}{\frac{k'_y aS}{(1-y)_{im}}(1-y)(y-y_i)}$$

Liq.

$$Z = \int_{x_2}^{x_1} \frac{L\,dy}{\frac{k'_x aS}{(1-x)_{im}}(1-x)(x_i-x)}$$

Gas (dilute)

$$\frac{V}{S}(y_1-y_2) = k'_y aZ(y-y_i)_m$$

Liq. (dilute)

$$\frac{V}{S}(x_1-x_2) = k'_x aZ(x_i-x)_m$$

Height of tower: using transfer units

$$H_G = \frac{V}{k'_y aS} \qquad N_G = \int_{y_2}^{y_1}\frac{(1-y)_{im}}{(1-y)(y-y_i)}\,dy$$

Dilute system

$$Z = H_G N_G = H_G\frac{y_1-y_2}{(y-y_i)_m}$$

⑥ Use of log-mean forms

$$(1-y)_{im} = \frac{(1-y_{AG})-(1-y_{Ai})}{\ln\frac{1-y_{AG}}{1-y_{Ai}}} \qquad (y-y_i)_m = \frac{(y_1-y_{i1})-(y_2-y_{i2})}{\ln\frac{y_1-y_{i1}}{y_2-y_{i2}}}$$

Adaptations

Pairs of symbols

k'_y	H_G	N_G	y_i
k'_x	H_L	N_L	x_i
K'_y	H_{OG}	N_{OG}	y^*
K'_x	H_{OL}	N_{OL}	x^*

1. **Absorption tower:**
- O.L. **above** E.L
- Minimal V'

Absorption

$y - y_i$
$x_i - x$
$y - y^*$
$x^* - x$

2. **Stripping tower:**
- O.L. **below** E.L
- Minimal L'

Stripping

$y_i - y$
$x - x_i$
$y^* - y$
$x - x^*$